DATS 598: Data Science Capstone

Journal Entries and Progress Documentation

Statistical Truth Detection System

Eric Crisp
University of Pennsylvania
ecrisp@upenn.edu

Monday 6th October, 2025

Contents

Journal Entry - Week 4 - Statistical Truth Detection System

Date: Monday 6th October, 2025

Objective

Synthesize the literature review and understand a path forward for the truth detection project. Also summarize current project standing.

To respond to the questions in the Week 3 Journal assignment:

- I did have a full meeting. The takeaways were about about implementing an analytical component to this project such as the influence of search on LLM output, like looping on fact generation until accurate as opposed to giving first response that may be incorrect.
- I mostly did literature review, environment configuration to have a stable development framework with various open source libraries, initial EDA with libraries and processes.
- My action items are to get the full infrastructure in place and begin working with the initial datasets, as well as produce initial NLP preprocessing steps.

My refelction on Phase 2 materials is as follows. The value of Pytorch in this project is impossible to understate. Pytorch, NLTK, spaCy, and HuggingFace are instrumental to not just the NLP components but also the ability to apply deep learning to the

Activities Completed

- Full stack development.
- Project scope and timeline iteration.
- Literature review.

Next Steps

- Develop NLP processing pipeline.
- Begin EDA on initial datasets (FEVER, Wikipedia, etc.)

The current tech stack (as planned, and currently built is this):

- Backend: Python, FastAPI, Uvicorn
- Frontend: JavaScript, React, Vite
- VCS/Deployment: Git, GitHub Actions for CI/CD, AWS

- Database: PostgreSQL, Pinecone
- NLP: SpaCy, NLTK, HuggingFace Transformers
- Containerization: Docker (Docker compose)
- EDA Packages: Jupyter Notebooks, Pandas, Matplotlib, Seaborn, NLTK, SpaCy, Transformers, Torch